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Excellent Electrolyte Wettability and High Energy Density of B₂S as Two-Dimensional Dirac Anode for Non-Lithium-Ion Batteries

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Abstract

Two-dimensional (2D) Dirac materials with ultrahigh electronic conductivity exhibit great promise for application as an anode material in non-lithium-ion batteries (NLIBs) with reduced conductive additive and binder as non-active material additive. Graphene is one of the most prominent 2D Dirac materials with high electrolyte wettability; however, it cannot be used as an anode material in NLIBs owing to its poor affinity towards metal ions such as Na, K, Ca, Mg and Al. In this work, we investigated the using of recently developed boron sulfide (B₂S) as a new light-weight 2D Dirac anode for NLIBs on the basis of first-principle calculations. We demonstrate that B₂S delivers excellent electronic conductivity and has a unique "self-doping" effect by forming S or B vacancies. Ultrahigh energy density of 2546 and 1223 mWh/g, a product of capacity and open circuit voltage referenced by standard hydrogen electrode potential (Cao et al. Nat. Nanotechnology. 2019, 14, 200-207), could be achieved for the B₂S anode in Na and K ion batteries, respectively, significantly larger than those of graphene. More importantly, the B₂S presents graphene-like wettability toward commonly used electrolytes in Na and K ion batteries, *i.e.*, the solvent molecules and metal salt, indicating excellent compatibility. Moreover, the minimum energy path (MEP) for Na and K ion diffusion on the B₂S surface shows an energy barrier of 0.19 and 0.04 eV, which indicates high ionic conductivity. Furthermore, a small contraction of the B₂S lattice upon ion intercalation has been observed due to the adsorption-induced corrugation of the electrode which offsets the lattice expansion. The results suggest that the B₂S electrode can be used as a light-weight 2D Dirac anode material with excellent energy density, desirable rate performance, and robust wettability toward the electrolytes.

Keywords: Sodium and potassium-ion batteries, electrolyte wettability, 2D Dirac anode B₂S, electrode corrugation, first-principle calculations

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1. Introduction

With the continued growth of the microelectronics industry and the electrification of transportation, the need for energy storage devices that deliver high capacity coupled with a fast charging/discharging rate has become more pressing. Lithium-ion batteries (LIBs), the most prevailing energy storage devices, have been widely used in portable electronic devices and fully electric vehicles due to their merits of a superior energy density, long cycle life, and relative environmental friendliness.¹ However, the large-scale application of LIBs has been strongly plagued by the limited accessibility and uneven distribution of Li in the earth's crust.² In this context, we have recently witnessed an increased research enthusiasm pertaining to non-lithium-ion battery (NLIBs), such as Na-ion,^{3,4} K-ion,^{3, 5} Ca-ion,³ Mg-ion,⁶ and Al-ion⁷ batteries, which benefit from the plethora of metal-ion reserves and low costs. Moreover, NLIBs share a similar operating mechanism to LIBs; hence, the technologies that have been developed for LIBs may be directly compatible with NLIBs or have broadly similar guiding principles for their development. However, they generally possess inferior electrochemical performances which include a low capacity, unsatisfactory energy density, and a short lifespan, owing to their larger ionic radius, higher standard potential, and heavier ionic weight compared with Li⁺.⁸

Many strategies have been proposed to address these issues, and among them, the development of novel electrode is an important one⁹⁻¹². Electrodes, core components of the battery, consist by cathode and anodes which provide enough space for itinerant metal-ions during the charging/discharging process. They determine not only the capacity of the battery but also dictate much of its rate performance.¹³⁻¹⁵ There has been significant process in achieving high performance from cathode materials, *i.e.*, the use of layered transition-metal oxides,¹⁶ manganese-based layered oxides,¹⁷ olivine,¹⁸ polyanion compounds,¹⁹ and framework materials,²⁰ however, advancements in the development of anodes lag much far behind. This is because many anodes which work well for LIBs perform poorly when directly used for NLIBs. To date, numerous anode materials have been reported as promising candidates for NLIBs, such as phosphorene,²¹⁻²² metal sulfides,²³ graphitic materials,²⁴ transition metal carbides (MXenes),^{3, 25} germanium selenide,²⁶ Si-based materials,²⁷ α - and β - tin and alloys²⁸. Unfortunately, many of them are semiconductors that suffer from low conductivity; this then requires the addition of conductive additives as well as binders to integrate the conductive additives, the active materials, and the current collectors. Since most of the binders in NLIBs are insulating and electrochemically inactive, their addition inevitably reduces the volumetric/gravimetric energy density of the batteries and introduces undesirable side reactions with the electrolyte.²⁹

Two-dimensional Dirac materials, a kind of materials with a linear band dispersion that exhibit ballistic charge transport and ultrahigh carrier mobility, are highly promising for anodes of NLIBS^{5, 30}. Their ultrahigh metallic conductivity and high specific area, which favor metal-ion adsorption, have stimulated great interest for application as binder-free 2D Dirac anodes. The most prominent 2D Dirac anode is graphene; however, pristine graphene is inactive toward metal ions. The introduction of lattice defects or extra edges could improve electrochemical metal-ion storage, but this breaks the Dirac cone in graphene and leads to a sizeable band gap with degenerated conductivity.³¹⁻³² Other 2D Dirac materials, such as silicene, germanene, and stanene exhibit better affinity to metal ions,

however, they suffer from a higher deadweight than graphene. Zhu et al. investigated the adsorption of Na on silicene and predicted a capacity of 954 mAh/g and a diffusion barrier of ~ 0.3 eV.³³ Recently, a chain-on-top adsorption model of Na on silicene has been reported by Xu et.al.³⁴ For germanene and stanene, based on the Bader charge analysis, Bohayra et.al predicted a lower capacity of ~ 369 and 226 mAh/g for Ge and Sn, respectively, as Na-ions battery anodes, because of their larger atomic mass.³⁰ Lu's group investigated creating a nanomesh of stanene to reduce the anode weight and improve its theoretical capacity.³⁵ These reports provide insights at the atomic level on the electrochemical storage of metal-ions in 2D Dirac anodes, and they have broadened the number of potential candidates for NLIBs. However, most previous theoretical works devoting to prediction of promising 2D Dirac electrodes concentrated only on their reversible capacity, while insufficient attention has been paid to their energy density.³⁶ Energy density is a product of the capacity and working voltage of a battery, and it represents a more appealing parameter for industrial application. It is noteworthy that previous works has generally focused on achieving high capacity while the contribution of the open circuit voltage (OCV) of the electrode has been ignored. For anode, only a high theoretical capacity coupled with a low OCV could yield a high energy density. More importantly, the wettability of the electrode by the electrolyte has drawn little attention; nonetheless this parameter is highly involved in cell packing and the electrochemical performance of the 2D Dirac anode.³⁷ Therefore, light-weight 2D Dirac anodes with a high conductivity, favorable electrolyte wettability, and excellent energy densities (a high capacity with a low OCV for the anode) is highly desired for NLIBs.

Recently, Zhao et al.³⁸ reported a brand new 2D Dirac material, B₂S, using a global structure search; it exhibited a light weight and favorable mechanical, thermal and dynamical stability. Li et al.³⁹ rationally designed an energy decomposition model to understand the Li adsorption behavior on graphene. This model was successfully applied to the B₂S system and demonstrated its high potential in electrode applications. However, investigations with regards to its feasibility as a 2D Dirac anode for NLIBs, *i.e.* the energy density, electrode wettability with electrolyte and solvent molecules, are still in their infancy. Herein, density functional theory (DFT) calculations and ab-initio molecular dynamics (AIMD) simulations were employed to scrutinize B₂S as a 2D Dirac anode material for NLIBs. The B₂S anode was shown to exhibit comparable wettability to graphene for commonly used NLIBs electrolytes (including the solvent molecule and the metal salt). Moreover, the emergence of vacancies in B₂S induces a unique self-doping phenomenon, where p-type doping could be obtained by the formation of B or S vacancies. Such "self-doping" increases the electronic conductivity of the material and is beneficial for reducing non-active material in the cell components. The electrochemical performances of the B₂S anode, such as the theoretical capacity, open circuit voltage, ion migration, volume expansion, and the energy density, have been systemically investigated. The stability of the electrode upon metal-ion intercalation was also assessed by AIMD simulations. This theoretical exploration provides helpful guidance for the development of new 2D Dirac anode materials and highlights the consideration of electrode wettability and energy density in searching for NLIBs electrodes.

2. Methods

All DFT calculations were carried out in the Dmol³ module⁴⁰, which employs numerical functions on an atom-centered grid as the atomic basis. The generalized gradient approximation (GGA)⁴¹ with the Perdew-Burke-Ernzerhof (PBE) was adopted as the exchange-correlation functional. For the description of Van der Waals interactions, the Grimme method was applied.⁴² All electronic including some relativistic effects were employed for the core treatment and the basis set used in all calculations was set as double-numerical polarization (DNP). To alleviate artificial interactions between the periodic images, a vacuum layer of ~20 Å was adopted. The orbital cutoff was set at 5.1 Å to obtain high-quality results. A 7×11×1 k-point grid was used to sample the Brillouin zone. The convergence criterion for geometry optimization of the energy, force, and displacement were 1.0×10^{-5} Ha, 0.002 Ha/Å, and 0.005 Å, respectively. A smearing value of 0.002 Ha was adopted in order to accelerate convergence. For metal-ion diffusion, the minimum energy paths were determined by the LST/QST methods with an RMS convergence of 0.002 Ha/Å. The calculation parameters were kept the same for all calculations to avoid systematic error.

The adsorption energies (E_{ad}) for the metal ion on the B₂S electrode are calculated as follows:

$$E_{ad} = E_{B_2S+M} - E_{B_2S} - \mu_M \quad (1)$$

where E_{B_2S+M} and E_{B_2S} are the total energy of the metal-loaded B₂S and the pristine B₂S, respectively; μ_M is the chemical potential of the metal atoms in their bulk state.

The theoretical specific capacity is obtained by

$$C = \frac{n_{max} z F}{M_{B_2S}} \quad (2)$$

where n_{max} is the maximum concentration of the adatom; $z = 1$; F is the Faraday constant (26801 mA h g⁻¹); M_{B_2S} is the weight of the B₂S electrode. It is important to select a true ground state structure at a given Na and K ion concentration to give a right estimation of the specific capacity and open-circuit voltage (OCV). In this work, we enumerate more than twenty possible configurations for a given Na and K ion concentration according to the different arrangement of metal-ion on B₂S surface. After geometry optimization, we select the structure with the lowest energy to represent the ground state structure whose lattice constants were fully optimized. Based on that, the specific capacity, OCV and mechanical properties were calculated.

The diffusivity of metal ions on B₂S could be estimated through the transition state theory⁴³: $D = \lambda^2 k$, where λ is the distance between adjacent diffusion sites and k is the kinetic rate constant.²²

According to the Eyring equation⁴⁴: $k = \frac{k_B T}{h} e^{\frac{-\Delta G_A}{k_B T}}$, where k_B is the Boltzmann constant, h represents the Planck's constant, T is the temperature and ΔG_A is the calculated Gibbs free energy of activation. We only consider the contributions from activation energy found by TS search, Zero-Point energy and vibrational entropy component because the volume change, rotational and translational entropy during this process are negligible. Accordingly, $\Delta G_A \approx E_{bar} + \Delta E_{ZPE} - T\Delta S_{vib}$, where the E_{bar} is the activation energy, ΔE_{ZPE} is the change of the Zero-Point Energy and ΔS_{vib} is the change of vibrational entropy

between the initial state and the transition state for metal ion diffusion.

3. Results and discussion

I. Atomic structure and electronic properties of pristine and defective B₂S

B₂S is an isoelectronic structure of graphene, and hence, it shares a graphene-like planar structure. However, unlike graphene, B₂S comprises two sulfur and four boron atoms with different radii in a hexagonal ring. This leads to a slightly deformed honeycomb structure and an orthogonal primitive cell. The optimized geometrical structure of the 1×2 B₂S sheet is shown in Figure 1a, with the unit cell highlighted by blue dashed rectangle. B₂S exhibits a planar configuration without any corrugation. Its lattice parameters were calculated to be $a = 9.14$ Å, and $b = 5.26$ Å. The calculated bond lengths of B-B (l_{B-B}) and B-S (l_{B-S}) are ~ 1.624 Å– 1.628 Å and ~ 1.815 Å– 1.841 Å, respectively, and is in good agreement with a previous report.³⁸ The energy band of the pristine material and the corresponding density of states under the PBE level are illustrated in Figure 1b. It can be seen that the two linear bands cross at the Fermi level, leading to the formation of a Dirac cone along the Γ -Y direction. The Dirac bands are solely composed of B-2p orbitals with tiny orbitals from the S atoms.

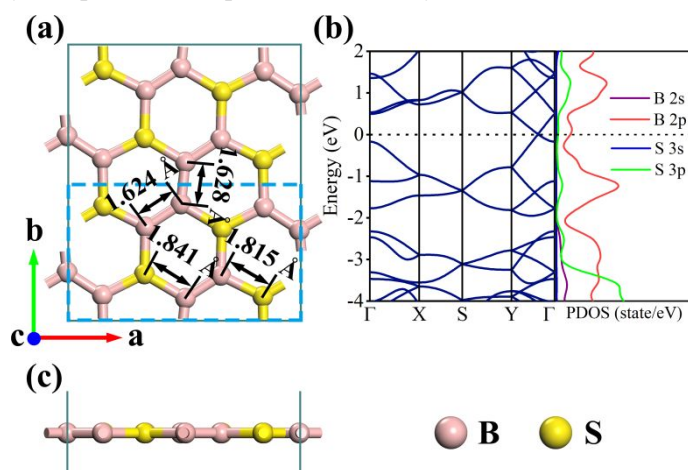


Figure 1. The atomic configuration of the 1×2 B₂S monolayer (a) and the electronic band structure as well as the partial density states (PDOS) (b). The unit cell of B₂S is marked by the blue dashed rectangle. The dotted line represents the Fermi level.

Vacancies are inevitable during electrode fabrication or with lattice vibrations at elevated temperature. These observations motivated us to investigate the formation of lattice vacancies in B₂S and consider its influences on the electrode conductivity. There are two types of vacancies in B₂S, *i.e.*, S and B, which can be simulated by removing a single B or S atom in a 2×3 B₂S supercell; this leads to a vacancy concentration of 1.4%. The optimized configurations for the defected B₂S are illustrated in Figure 2. The emergence of a vacancy in B₂S yields an obvious local atomic reconstruction. The two boron atoms close to the vacancy move closer, accompanied by the formation of a pentagon-like structure (Figure 2a and 2d). The formation energy (E_f) for S vacancy is 9.33 eV while those for the B vacancies, B_I and B_{II} in Figure 2d and 2g, are 8.61 and 8.48 eV, respectively. This indicates that the formation of S vacancies is a little harder than B vacancies. E_f is given by $E_f = E_{\text{defect}} + E_a - E_{\text{perfect}}$, where E_{defect} , E_a , and E_{perfect} are the total energy of the defective B₂S, the isolated missing atom, and

the perfect B₂S, respectively. The calculated E_f values are higher than that of graphene (7.7 eV)⁴⁵, implying the robust bonding between B-S or B-B in B₂S. In contrast to graphene, where the emerging vacancy results in the magnetism, B₂S with S or B vacancies exhibit a non-magnetic ground state. Moreover, it is well known that the formation of vacancies in graphene destroys the delocalized π electron network, inducing obvious electron scattering and a deterioration of the metallic conductivity. We observed a “self-doping” in B₂S either by B or S vacancies in addition to a small energy gap as shown in Figure 2c, 2f, and 2i; here, the valence band crosses over the Fermi level and this is indicative of a p-type doping. Such p-type doping is beneficial for achieving high intrinsic conductivity without the introduction of extra conductive additive and thus holds great potential to serve as binder-free anode materials. The “self-doping” phenomenon arises from the electron deficiency caused by the removal of B or S. To facilitate the experimental investigation of vacancy defects in B₂S, we simulated their STM images as shown in Figure 2b, 2e, and 2h.

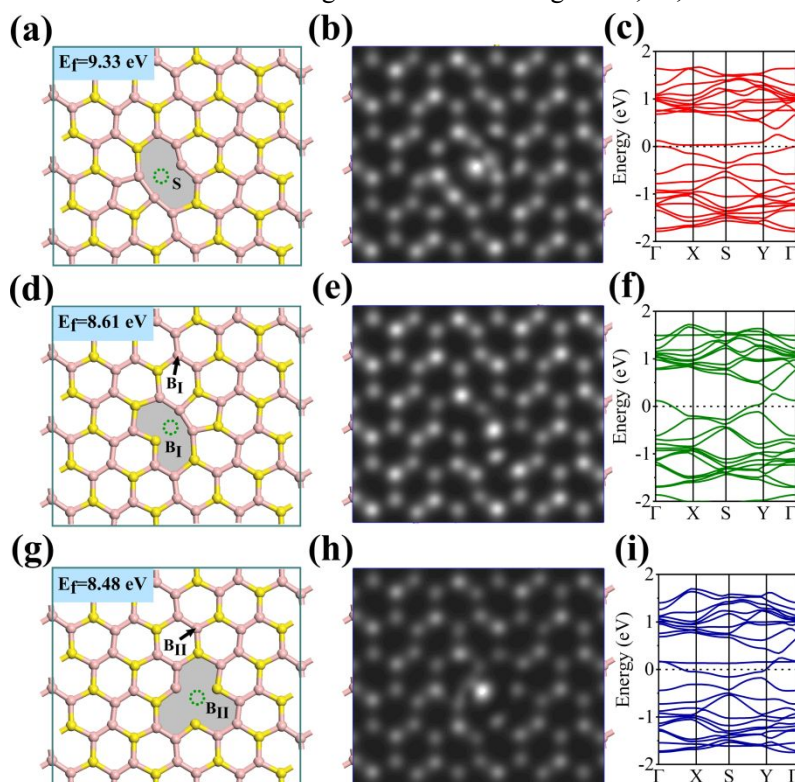


Figure 2. The relaxed configurations of the 2×3 B₂S supercell with a single S vacancy (a), B_I vacancy (d), and B_{II} vacancy (g) and the simulated STM profiles for the S vacancy (b), B_I vacancy (e), and B_{II} vacancy (h). Their electronic band structures are given in (c), (f), and (i), respectively. The vacancy is marked in light grey for easy viewing and the green dotted circles represent the missing atom. The STM bias was set to be 1.5 V.

II. Na, K, Ca, Mg and Al ion adsorption on B₂S electrode

The energy density, a key parameter for NLIBs applications, is in proportion to the product of the capacity and voltage.³⁶ Larger adsorption energies (E_{ad}) are advantageous in preventing the formation of metallic dendrites and are beneficial for achieving a higher capacity. However, a large E_{ad} also

leads to a high discharge plateau in the anode. This reduces the total voltage output of the battery system, which is determined by the voltage difference between the average charge and the discharge plateau of the cathode and anode materials. Therefore, for a high-performance NLIBs anode, a delicate affinity towards metal ion with a moderate E_{ad} is required.

Towards this goal, we started by investigating the adsorption of Na, K, Ca, Mg and Al on the B₂S electrode surface. To determine the most energetically favorable adsorption site for the metal ion on the B₂S electrode surface, several types of candidates, including H sites (H1 and H2), B sites (B1, B2, B3, and B4), and T sites (T1, T2, and T3) were considered as shown in Figure 3a. The most energetically favorable one for Na, K, and Mg is H1, whereas Ca and Al prefer to adsorb on the T2 site. The calculated E_{ad} of the ground state configuration is given in Table 1 (E_{ad} values for other adsorption sites can be found in Table S1). It can be seen that the B₂S electrode shows negative E_{ad} values only for the Na and K atoms, while those for the Ca, Mg and Al atoms are all positive. According to the definition, a positive E_{ad} indicates that the metal atoms prefer to form metallic dendrites or metallic clusters rather than residing on the B₂S monolayer separately. Thus, B₂S cannot be applied to the anode for Ca, Mg and Al ion batteries. Intriguingly, the E_{ad} for Na and K are -0.29 and -0.76 eV, respectively. This is large enough to avoid metallic dendrite formation, yet with an adequate, moderate strength to obtain the desired voltage. For adsorption, a charge transfer (Δq) of 0.47 and 0.59 e for Na and K, respectively, was determined by Hirshfeld analysis, which is in line with their E_{ad} trend shown in Table 1. We also conducted Bader charge analysis as shown in Figure S1, which yield a little larger charge transfer of 0.80 and 0.85 e for Na and K adsorbed on B₂S anode owing to their different definition of interatomic boundary⁴⁶.

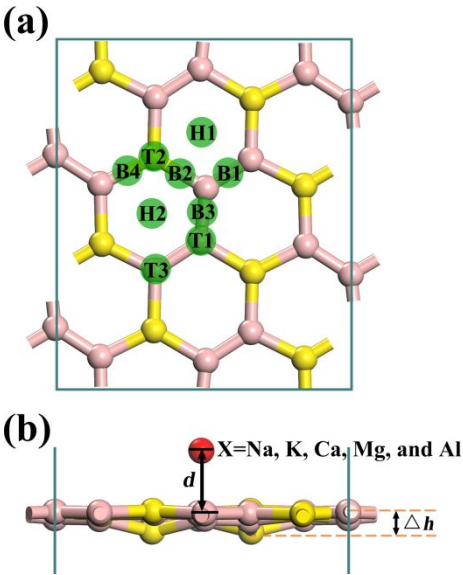


Figure 3. (a)-(b) Top and side views of the considered adsorption sites for single metal atom (X, X= Na, K, Ca, Mg and Al) on the B₂S surface. H, B, and T represents that X adsorbs above the center of the hexagonal rings, on top of the B or S atom, and above the B-B or B-S bridge site of B₂S, respectively. d is the adsorption height and Δh refers to the corrugation height of B₂S after adsorption.

Table 1 The calculated E_{ad} , transfer charge (Δq), adsorption height (d), and corrugation height (Δh) of the B₂S electrode with the adsorbed metal ion.

Metal ions	E_{ad} (eV)	Δq (e)	d (Å)	Δh (Å)
Na	-0.29	0.47	1.92	0.56
K	-0.76	0.59	2.40	0.43
Ca	0.08	0.56	1.84	0.93
Mg	0.90	0.32	1.82	0.69
Al	1.33	0.20	1.54	0.88

The charge density differences between the Na and K ions and the B₂S electrode are calculated to shed light on the adsorption mechanism as shown in Figure 4a-b. The orange and light blue areas present the electron accumulation and depletion, respectively. An obvious electron depletion emerged around the Na and K atoms with electron accumulation on the B₂S electrode, which was in agreement with Hirshfeld charge analysis. However, the transferred electrons accumulated locally on the hexagonal ring beneath the adsorbed ions, yielding an electrostatic repulsion between B-B and B-S. Thus, the l_{B-S} increased from 1.816 Å (1.841 Å) to 1.857 Å (1.887 Å) for Na and 1.841 Å (1.868 Å) for K ion adsorption, and the l_{B-B} decreased from 1.624 Å to 1.616 Å for Na and 1.615 Å for K ion adsorption. Simultaneously, the two S atoms nearby protruded from the B₂S plane, yielding a corrugation height (Δh) of 0.56 and 0.43 Å. The partial density of states (PDOS) of on the B₂S electrode with adsorbed Na and K ions is shown in Figure 4c-d. It reveals that the adsorption of metal ions does not change the inherent metallic property of the B₂S electrode, where the B-2p, S-3p, Na-3s, and K-4s orbitals were dominant at the Fermi level. Moreover, an upward movement of the Fermi level was observed as the transferred electrons from the metal ions occupied the conduction band of B₂S. This increased the carrier concentration in the B₂S electrode, which is advantageous for achieving high electrode conductivity.

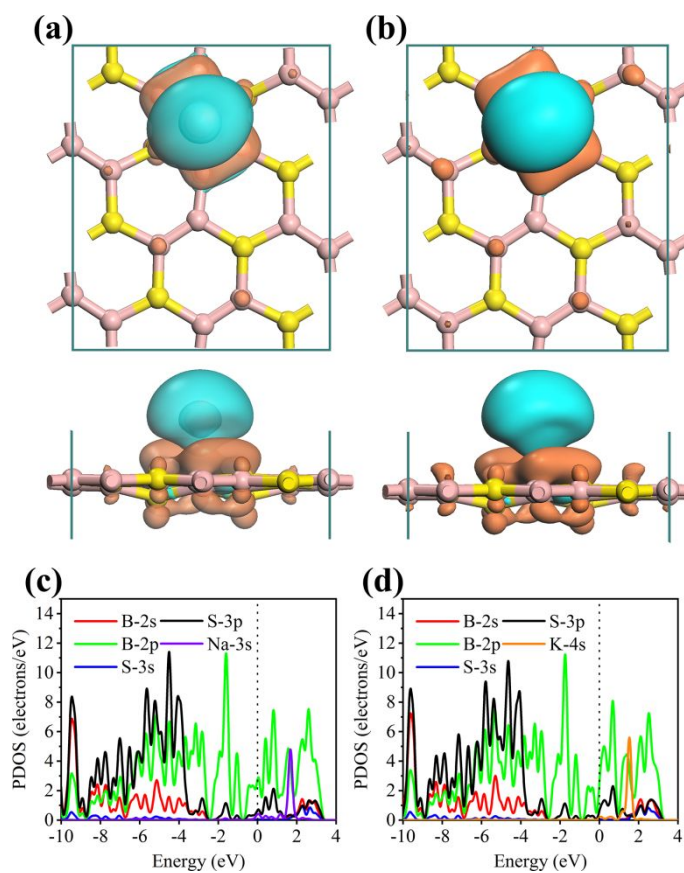


Figure 4. Plots of the charge density difference (ρ) for the Na ion (a) and the K ion (b) on the B₂S surface and the corresponding partial density of states (PDOS) for the Na ion (c) and the K ion (d). ρ is defined as: $\rho = \rho_{B_2S+Metal} - (\rho_{B_2S} + \rho_{Metal})$, where $\rho_{B_2S+Metal}$, ρ_{B_2S} , and ρ_{Metal} are the electron density of the adsorption system, the B₂S electrode, and the metal atom, respectively. The orange and light blue areas refer to the electron accumulation and depletion, respectively. The Fermi level is indicated by a dotted line and is set at zero.

III. Wettability of typical electrolytes in B₂S electrode

The wettability, which is ability of a liquid to spread over the solid surface, is of great important in cell packing and affects the electrochemical performance of the active electrode. Insufficient or non-uniform wetting results in irregular reaction at the electrode, inhomogeneous distribution of current density and the formation of an unstable solid electrolyte interface film. This degenerates the cell performance, *i.e.* the energy density, and can even result in the safety hazards by forming localized lithium plating on the anode.³⁷ The experimental characterization of the wettability is usually performed by measuring the contact angle. The most fundamental physical variable determining the contact angle are the solid-liquid interactions which can be easily estimated by the adsorption energy (E_{ad}) of the liquid molecules to the solid surface. In the case of the anode material, a higher E_{ad} for the electrolyte molecules (*i.e.*, solvent molecule and metal salt) on the electrode generally leads to better wettability and *vice versa*. The wettability of some commonly used electrolytes in Na and K ion batteries on the B₂S electrode surface were considered (*i.e.*, BTFE, DEC, DMC, EC, FEC, and PC for

solvent molecule⁴⁷ and NaClO₄, NaFSI, NaPF₆, KPF₆, KFSI, and KTFSI for metal salt⁴⁷⁻⁵¹). To determine the most stable adsorption site for the electrolytes on B₂S, several different adsorption configurations were checked and the most energetically favorable one is shown in Figure 5 and Figure S4. For comparison, electrolyte adsorption on graphene electrode has also been investigated and the most stable adsorption structure is shown in Figure S2 and S3. The calculated E_{ad} and adsorption heights are listed in Table 2. Generally, all the considered solvent molecules exhibit favorable adsorption strength over the B₂S surface. They have an E_{ad} in the range of $-0.40 \sim -0.69$ eV and most of them are comparable to those of graphene ($-0.54 \sim -0.87$ eV) except for BTFE. In the case of BTFE, its E_{ad} on B₂S is much smaller than the graphene, indicating inferior wettability on the former. Moreover, the binding of the solvent molecules with B₂S occurs through van der Waals interactions, which is further evidenced by the larger adsorption height. Similar observations have also been found for metal salt as shown in Table 2. However, the metal salt exhibited much stronger binding than solvent with E_{ad} values of $-0.92 \sim -1.32$ eV, and hence their addition could facilitate the wetting process. Notably, for Na-containing salt, B₂S showed an even better affinity toward NaClO₄ and NaFSI than graphene, whereas, K-containing salt (KPF₆, KFSI, and KTFSI) had reduced adsorption to B₂S. To this end, B₂S could provide graphene-like wettability towards most of the commonly used solvents when used as the anode in Na and K ion batteries.

Electrolyte stability after contacting with the electrode is another essential issue for practical application. The stability can be estimated by the energy level alignment of the electrode's Fermi level with the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) of the electrolyte molecules.⁵² It has been demonstrated that graphene is stable with currently used electrolytes. The work function of B₂S is 4.57 eV, a little smaller than that of graphene (4.63 eV), implying its stability when in contact with the electrolytes. The electronic band structure of the B₂S anode with metal salt and adsorbed solvent are shown in Figure 5 and Figure S4. Notably, only tiny energy gap open in B₂S electrode due to the electronic scattering from electrolytes.

Table 2 The calculated adsorption energy (E_{ad}) and corresponding adsorption height (ΔH) of the electrolytes, BTFE, DEC, DMC, EC, FEC and PC for the solvent molecules and NaClO₄, NaFSI, NaPF₆, KPF₆, KFSI, and KTFSI for the metal salt on B₂S electrode, as well as adsorption energy difference (ΔE_{ad}) between graphene and B₂S.

Electrolytes		Graphene		B ₂ S		ΔE_{ad} (eV)
		E_{ad} (eV)	ΔH (Å)	E_{ad} (eV)	ΔH (Å)	
Solvent molecule	BTFE	-0.66	2.53	-0.40	3.09	-0.26
	DEC	-0.87	2.44	-0.69	2.57	-0.18
	DMC	-0.54	2.59	-0.44	2.71	-0.10
	EC	-0.56	2.36	-0.51	2.30	-0.05
	FEC	-0.60	2.34	-0.49	2.47	-0.11
	PC	-0.69	2.28	-0.63	2.34	-0.06
metal salt	NaClO ₄	-1.09	2.61	-1.14	2.56	0.05
	NaFSI	-1.24	2.73	-1.32	2.23	0.08
	NaPF ₆	-1.20	2.71	-0.96	2.65	-0.24

	KPF ₆	-1.39	2.75	-0.92	2.76	-0.47
	KFSI	-1.61	2.33	-1.10	2.27	-0.51
	KTFSI	-1.56	2.87	-0.92	2.75	-0.64

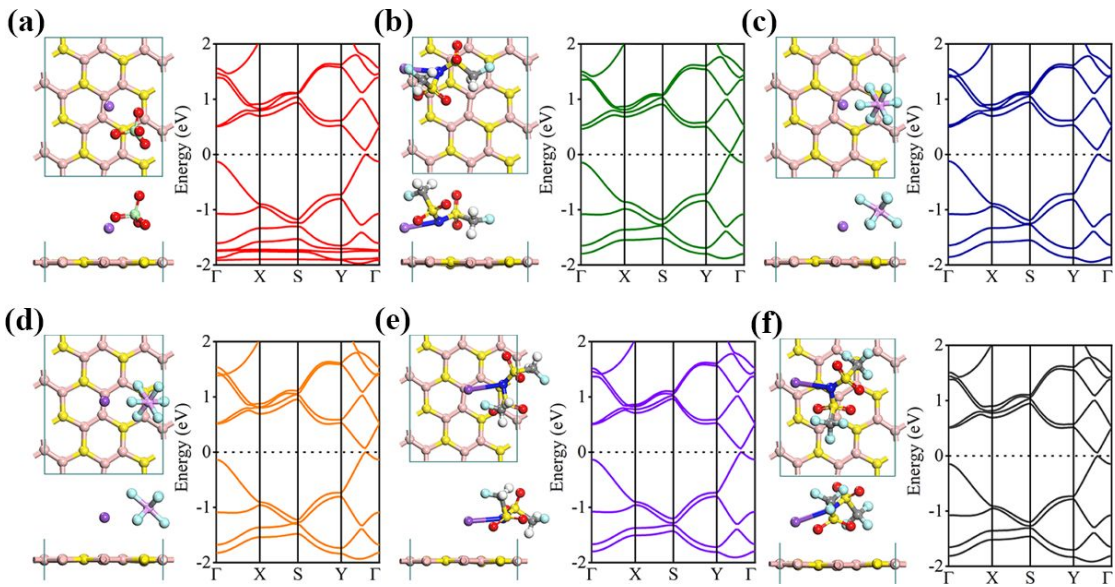


Figure 5. Band structures of the B₂S electrode with adsorbed NaClO₄ (a), NaFSI (b), NaPF₆ (c), KPF₆ (d), KFSI (e), and KTFSI (f) on the B₂S surface. The Fermi level is indicated by dotted line and set at zero.

IV. Diffusion of Na and K ion on the B₂S surface

The rate performance of the NLIBs is closely related to the mobility of the carriers and diffusion barrier for metal ions on the electrode surface. Electron transport (electronic conductivity) in the external circuit is significantly faster than the transport of metal ions (ionic conductivity) in the active electrode material, the electrolytes, and the separator. Moreover, metal ion diffusion in a liquid is also more favorable than in the solid phase. Therefore, the rate performance of cell is mainly dependent on the ion diffusion resistance in the electrode material. Based on that, the diffusion resistance of the Na and K ions on B₂S surface has been investigated by calculating their migration barrier (E_{bar}). Compared with graphene and silicene, the structural symmetry of B₂S is quite different. In the case of B₂S, two types of hexagonal rings coexist, *i.e.*, one hexagonal ring with two meta-S atoms and the other with two para-S atoms, which yields an orthogonal lattice and varied diffusion paths for metal ions. Consequently, two typical diffusion pathways were selected with the metal ions migrating from one energetic site to the next nearest one as illustrated in Figure 6a and 6d. The energy profiles corresponding to the two diffusion pathways are depicted in Figure 6b, 6c, 6e, and 6f. Notably, the E_{bar} for the Na ion crossing over the bridge sites of B-B and B-S are relatively close (0.19 eV vs. 0.21 eV, respectively), which are in agreement with Li's report.³⁹ However, the diffusion path for K ions on the B₂S are quite different, where a metastable adsorption was involved in the diffusion. This leads to anisotropic diffusion barriers for Path II (0.04 eV) and Path I (0.19 eV). Thus, the rate-limiting step in Path I is the K ion diffusion crossing over the bridge site formed by the B-B bond. Accordingly, the

migration of K ions along Path II is 2.25×10^2 times faster than along Path I at room temperature, exhibiting obvious direction-dependent diffusion. For comparison, E_{bar} for Na is smaller than borophene (0.33 eV),⁵³ black phosphorene (0.38 eV),⁵⁴ and Y_2C (0.39 eV)⁵⁵ and comparable to TiC_3 (0.18 eV),⁵⁶ and TiS_2 (0.22 eV),¹⁴ as shown in Table 3. With regards to K-ion, its E_{bar} is comparable to TiS_2 (0.09 eV),⁵⁷ Ti_3C_2 (0.10 eV),³ and graphene (0.12 eV),⁵⁸ while much smaller than hydrogen boride (0.32 eV).⁵⁹ We further calculated the diffusivity of Na and K on B_2S anode at each rate-limiting step, which were $2.33 \times 10^{-6} \text{ cm}^2/\text{s}$ and $1.82 \times 10^{-5} \text{ cm}^2/\text{s}$, respectively. These values are comparable with blue phosphorene (*i.e.*, $7.2 \times 10^{-5} \text{ cm}^2/\text{s}$ for Na and $8.58 \times 10^{-5} \text{ cm}^2/\text{s}$ for K)²². In addition, the diffusivity of Na on B_2S anode is much larger than its diffusion in hard carbon ($10^{-9} \sim 10^{-13} \text{ cm}^2/\text{s}$)⁶⁰. As for K, its diffusivity is in the same magnitude of Li diffusing along the surface of bilayer graphene.⁶¹ Therefore, the B_2S exhibits strong potential to serve as a high rate anode given both its high electronic conductivity and ionic conductivity.

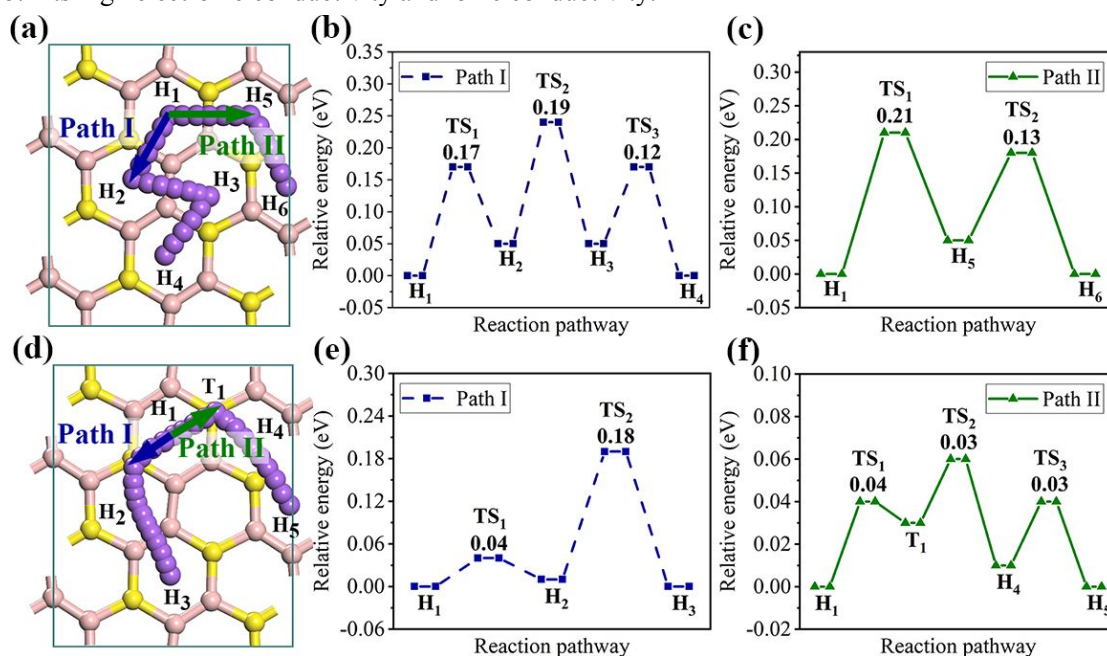


Figure 6. Schematic illustration of the diffusion pathway for the Na (a) and K-ions (d) on the surface of the B_2S electrode. The corresponding energy profiles for Na-ion diffusing along path I (b) and path II (c), and those for the K-ion along Path I (e) and Path II (f) are given.

V. Theoretical capacity, open-circuit voltage and electrode stability of B_2S

Next, we investigated the theoretical capacity, OCV, and the corresponding energy density of the B_2S anode for application in Na and K ion batteries. To mimic the charging process, metal ions were added in a stepwise manner onto B_2S surface, from which the voltage vs ions concentration in anode could be derived. In principle, the metal atoms could occupy both sides of the B_2S anode until the adsorption energy (E_{ave}) becomes positive or convergent, where the maximum storage capacity is achieved. The E_{ave} was defined as $E_{\text{ave}} = (E_{\text{B}_2\text{S} + x\text{M} + y\text{M}} - E_{\text{B}_2\text{S} + x\text{M}} - y\mu_{\text{m}}) / y$, where $E_{\text{B}_2\text{S} + x\text{M}}$, $E_{\text{B}_2\text{S} + x\text{M} + y\text{M}}$, and μ_{m} are the total energy of the B_2S anode with x and $x + y$ metal ions, and the chemical potential of the metal atoms in their bulk state, respectively. Thus, the electrode potential of the B_2S anode upon

metal-ion intercalation could be calculated by: $V = -E_{\text{ave}} / ne$, where n is the number of charges transferred during the half-cell reaction ($n = 1$ for both Na and K). Finally, the B_2S electrode is able to host up to 16 Na and 8 K, which have a chemical stoichiometry of $\text{Na}_2\text{B}_2\text{S}$ and KB_2S , respectively. The fully metalized phase satisfy the requirements proposed by Zhao et. al, that is, with negative adsorption energy for metal-ion adsorption, without metal-ions pushing out of the anode, and without irreversible electrode deformation⁶². Accordingly, the B_2S anode exhibits a theoretical capacity of 998 and 499 mA h g^{-1} for the Na and K ion batteries, respectively. The capacity calculated in our work is a little smaller than that of Li et al. due to the different capacity calculation methods.³⁹ Herein, we have considered the dynamic adsorption process of metal ions on a preadsorbed electrode surface, where the adsorption process was considered as finished (a maximum capacity achieved) when the E_{ave} of the post adsorbed metal ions becomes positive.

To capture the voltage evolution of Na and K ion adsorption at each ion concentration and investigate the stability of the related intermediated phases, we computed the formation energy (E_f) of $\text{Na}_x\text{K}_p\text{B}_2\text{S}$ referencing to the two end structure, *i.e.*, B_2S and $\text{Na}_p\text{B}_2\text{S}$ (p is 2 for Na and 1 for K). E_f is given by: $E_f = E_{\text{Na}_x\text{K}_p\text{B}_2\text{S}} - \left[\frac{x E_{\text{Na}_p\text{B}_2\text{S}} + (p-x) E_{\text{B}_2\text{S}}}{p} \right]$, where $E_{\text{Na}_x\text{K}_p\text{B}_2\text{S}}$ is the total energy of B_2S with x Na or K ions.⁶³⁻⁶⁵ Accordingly, the thermodynamically stable phases can be identified from the energy convex hull as shown in the inset of Figure 7. The convex hull for both Na and K presents only one stable intermediate phase, that is NaB_2S and $\text{K}_{0.5}\text{B}_2\text{S}$, respectively. Thus, the intercalation potential of Na and K has to be computed by considering only the two end points and the phase laying in the hull. In Figure 7, the voltage evolution at each ion concentration is plotted as well as the corresponding atomic configurations. Two main voltage plateau exist for Na and K. In the case of $\text{Na}_x\text{B}_2\text{S}$, the first voltage plateau is 0.57 V vs Na/Na^+ , which refers to the Na ion concentration increasing from $x = 0$ to 1 in $\text{Na}_x\text{B}_2\text{S}$ as shown in Figure 7a. The second voltage plateau of 0.06 V comes from $x = 1$ to 2. KB_2S demonstrated a similar variation but with higher potential values from 1.00 to 0.18 V, as shown in Figure 7b. The voltage fades on increased metal ion adsorption occurred in B_2S anode is advantageous because it would increase the working voltage of the battery when coupled with a cathode material. To this end, the average OCVs are calculated to be 0.32 and 0.59 V for the Na and K ion batteries, respectively.

Previous work on 2D anode materials mainly concentrated on a half cell, and the effects of the working voltage on the energy density of the anodes usually got little attention. Having determined both the capacity and the average OCV, we can provide a meaningful estimation of the energy density for the B_2S anode by using the method proposed by Cao et.al., where standard hydrogen electrode, SHE, is employed as the universal cathode reference potential.³⁶ Such that, we could forecast the electrochemical performance of B_2S anode in terms of the energy density calculated from the capacity and the relative potential against the SHE reference of a half-cell set-up. Accordingly, the energy density of the B_2S anode for Na and K ion battery are calculated to be 2715 and 1223 mWh/g , respectively. As shown in Table 3, the reported energy density for the Na ion anode is much higher than the reported values of some well-known 2D electrode material, such as silicene^{30, 34} (2385

mWh/g), borophane⁶³ (1517 mWh/g), and borophene⁶⁶ (2301 mWh/g), but it is smaller than that of hydrogen boride⁶⁷(3413 mWh/g). For the K ion anode, its energy density is larger than that of $\text{Ti}_3\text{C}_2\text{O}_2$ (459 mWh/g) and TiS_2 ⁵⁷ (805 mWh/g), and smaller than borophene⁶⁶(2024 mWh/g) and hydrogen boride⁵⁹(3186 mWh/g). It should be noted that two different DFT calculation methods were reported in calculating the anodic theoretical capacity depending on whether the mass of metal ions was included or not.²⁵ Because most of the anode capacity in references are based on the neglect of the metal-ion mass, *i.e.*, the capacity of the most famous graphite anode for lithium-ion batteries, all the capacity and energy density data referenced in this article were thus based on formula (2) for the sake of comparison. The results including the metal-ion mass are provided in Table S2.

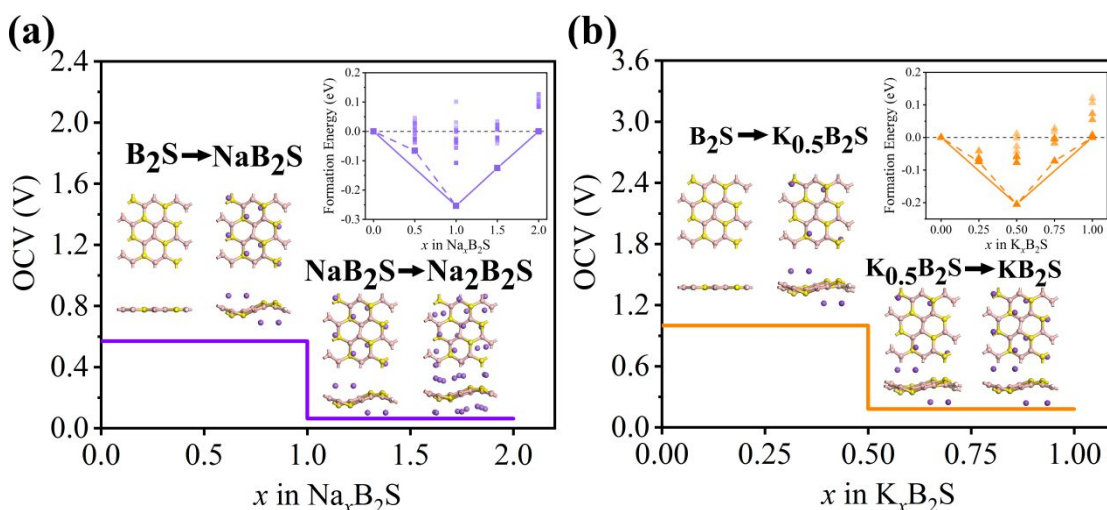


Figure 7. The calculated OCVs of B_2S anode for (a) Na and (b) K adsorption at different stage. The inset figures give the formation energy of $\text{Na/K}_x\text{B}_2\text{S}$ per formula unit. The configuration with formation energy lying on the tie line of the convex hull is also given as insets.

Table 3 A comparative summary of the energy densities and diffusion barriers of different anode materials for Na and K ion batteries determined from the literature. * These data were converted in calculating energy density for comparison, where the mass of metal ions was considered upon calculating the anode capacity in the original reports.

Anode materials	Na		K	
	E_{bar} (eV)	Energy density (mWh/g)	E_{bar} (eV)	Energy density (mWh/g)
Graphite ^{5, 68}	—	63	—	773
Silicene ^{30, 34}	0.12	2385	—	—
Stanene ³⁰	0.14	—	—	—
Stanene nanomesh ³⁵	0.15	512	—	—
Borophane ⁶³	0.09	1517	—	—
Borophene ⁶⁶	0.24	2301*	0.12	2024*
Hydrogen boride (BH) ^{59, 67}	0.32	3413	0.32	3186
MoS_2 ⁶⁹⁻⁷⁰	0.28	299*	—	466
MoC_2 ⁷¹	0.02	379	—	—
VS_2 ⁷²⁻⁷³	0.09	405	0.06	951
NbS_2 ¹⁴	0.07	713*	—	—
TiS_2 ^{14, 57}	0.22	1011*	0.09	805

Ti ₃ C ₂ ³	0.10	915	0.10	459
Y ₂ C ⁵⁵	0.01	1579	—	—
Black phosphorene ⁷⁴	0.04	920	—	—
B ₂ S	0.19	2715	0.04	1223

In many anode materials, sodiation and potassiation were found to expand the lattice volume, which even resulted in the electrode fracture.⁷⁵ To address the volume and mechanical properties changes of B₂S upon increased Na and K ion adsorption, their in-plane stiffness (*C*) was calculated as shown in Figure S5. Of importance to note is that obvious electrode corrugations could be observed upon Na and K ion charging in the B₂S anode. Benefiting from this, the lattice expansion is mostly offset and lattice contraction appeared. As indicated in Figure S5, the lattice contraction is $\varepsilon = -0.07$ for Na₂B₂S and -0.02 for KB₂S with $\varepsilon = (l - l_0) / l_0$, where *l* and *l*₀ are the pristine lattice constant and that of fully metalized anode, respectively. This was also evidenced by the increased wrinkle heights from zero in the pristine B₂S to 2.09 and 1.52 Å for the Na and K covered electrodes. A similar transition from a planar to a buckled structure was also observed in Si₂BN.⁶⁵ Accordingly, the in-plane stiffness reduced significantly from 139 N/m to 47 N/m for Na and 85 N/m for K in the zigzag direction. However, such reduced in-plane stiffness comes mainly from the unique adsorption-induced anode bending along this direction. In the armchair direction, the in-plane stiffness are nearly maintained. With low in-plane stiffness and excellent flexibility, B₂S is capable of enduring large strains without breaking. The band structures of B₂S with increased Na and K ion concentration are shown in Figure S6. As for Na adsorption, the metallic conductivity increase with the Na ion concentration with more electronic states crossing over the Fermi level. Similar observation was found for K adsorption while the system become semiconductive at KB₂S, which indicates the need of conductive additive.

AIMD simulations at 300 K for 3-6 ps were performed to estimate the thermal stability of B₂S anode at higher Na and K ions. The energy evolution as well as snapshots of Na_xB₂S and K_xB₂S after AIMD simulations are depicted in Figure S7. During the simulations, the electrode structure remained complete without any observed structure collapse. The fluctuation of the relative energy of the structures are also very small (less than 0.02 eV) except for Na₂B₂S, indicating their high thermodynamic stability at room temperature. As shown in Figure S7c, a Na ion escapes from the electrode surface after 6ps simulation, which yields a practical energy density of 2546 mWh/g at room temperature. The reversibility of the electrode was verified by removing the metal ions and performing AIMD simulations on the remaining B₂S (Figure 8). The corrugated electrodes were partially recovered to flatter configurations after 5 ps. Upon further optimizing their lattice, their configurations were fully recovered to their initial state, showing favorable cyclical stability.

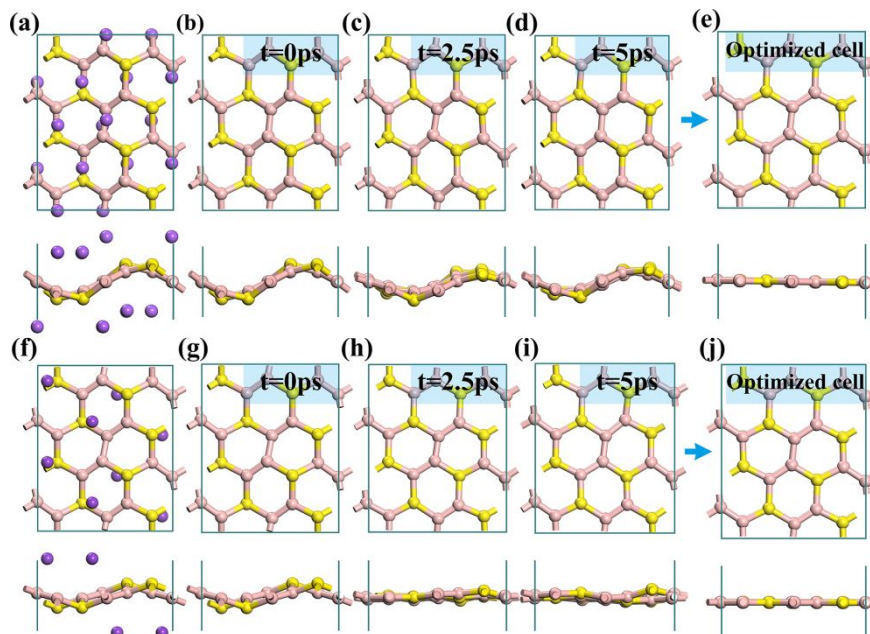


Figure 8. AIMD simulations (300K) of the B₂S electrode recovery process with fully removed (a-e) Na and (f-j) K ions.

In the last part, we discuss the potential synthesis routes for the B₂S anode. Recently, the rapid advancement of substrate-dependent epitaxy technology makes the synthesis of new two-dimensional materials possible⁷⁶, *i.e.*, various borophene polymorphs which have no analogous bulk layered allotropes. We expect that the B₂S monolayer, sharing a similar configuration to hexagonal borophene that was synthesized recently on Al (111)⁷⁷, could also be directly fabricated on a proper substrate or through a proper post-treatment toward the synthetic hexagonal borophene. To prove the latter hypothesis, we simply calculated the formation energy of B₂S referenced to a stepwise substitution reaction of hexagonal borophene by sulfur atoms: $E_{\text{formation}} = E_{\text{h-B-S}_x} + E_{\text{S}} - E_{\text{h-B-S}(x-1)} - E_{\text{B}}$, where $E_{\text{h-B-S}_x}$, $E_{\text{h-B-S}(x-1)}$, E_{S} and E_{B} is the total energy of 1×3 hexagonal borophene with x , $x - 1$ B atoms substituted by S atom, and the total energy of single S and B atom. Figure 9a schematically illustrates the proposed substitution process and the corresponding $E_{\text{formation}}$ are provided in Figure 9b. These reactions present obvious substrate-dependent, where all considered substrates present positive $E_{\text{formation}}$ except for MoS₂. On MoS₂ substrate, the fabrication of B₂S through a step-by-step sulfur substitution are all energy favorable, which might be attributed to the favorable bonding between the S atom in B₂S and the Mo atom in MoS₂ substrate.

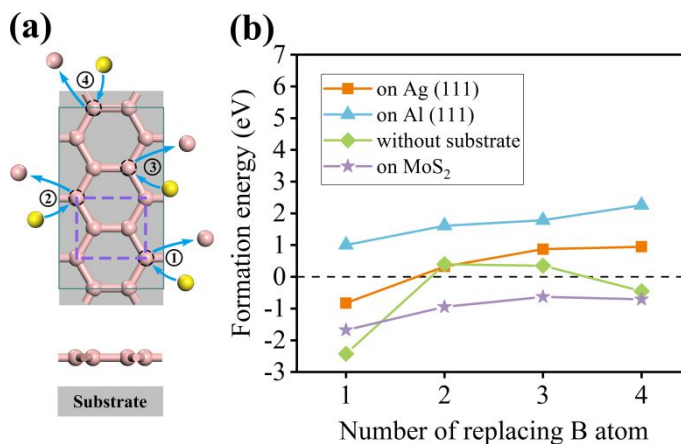


Figure 9. (a) Schematic illustration of the formation process of B₂S through a step-by-step substitution reaction of the 1×3 hexagonal borophene by S atoms, and (b) the corresponding formation energy profile on Ag (111), Al (111) and MoS₂ substrate.

4. Conclusion

In summary, we systematically investigated the potential of the B₂S monolayer as a 2D Dirac anode material for NLIBs based on first-principles calculations. The B₂S exhibited robust B-B and B-S bonding; a high vacancy formation energy of 9.33 eV was determined for the S vacancy, and 8.61 and 8.48 eV for B_I and B_{II}, respectively, which is larger than that of graphene. The formation of the B or S vacancy results in a “self-doping” effect, which leads to high electronic conductivity for the B₂S anode. As an anode material for Ca, Mg and Al ion batteries, metallic dendrites or clusters can be formed in the anode, which causes with safety hazards because of their positive E_{ad} . However, the B₂S surface has moderate adsorption strength for the Na and K ions, yielding an ultrahigh energy density of 2546 and 1223 mWh/g, respectively. In addition to its high electronic conductivity, the B₂S anode also has a high ionic conductivity for the Na and K ions with a diffusion barrier of 0.19 and 0.04 eV, respectively. More importantly, the B₂S anode exhibits excellent wettability towards most of the widely used electrolytes in the Na and K-ion battery systems, which is comparable to the prevailing graphene anode. Moreover, a negative expansion could be observed in the metalized B₂S electrode due to competition between adsorption-induced electrode corrugations and lattice expansion. The high energy density, excellent electrolyte wettability, tiny electrode expansion/contraction, and high electronic and ionic conductivity suggest that B₂S is a promising anode material for Na and K ion batteries.

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Conflicts of interest

There are no conflicts to declare.

Supporting information

The calculation method for Bader charge; calculated E_{ad} values for metal atoms on B₂S surface at all considered adsorption sites; the Bader charge for Na and K adsorbed anode; top and side view of most energetically favorable configurations of solvent molecule (BTFE, DEC, DMC, EC, FEC and PC) and metal salt (NaClO₄, NaFSI, NaPF₆, KPF₆, KFSI and KTFSI) adsorption on graphene; band structures and configurations of solvent molecule adsorbed on B₂S; the band structure and mechanical properties change at higher Na and K ion concentration; the AIMD results of Na and K covered B₂S anode at room temperature.

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Reference

1. Dunn, B.; Kamath, H.; Tarascon, J.-M. J. S., Electrical Energy Storage for the Grid: A Battery of Choices. *Science* **2011**, *334*, 928-935.
2. Yabuuchi, N.; Kubota, K.; Dahbi, M.; Komaba, S., Research Development on Sodium-ion Batteries. *Chem. Rev.* **2014**, *114*, 11636-11682.
3. Er, D.; Li, J.; Naguib, M.; Gogotsi, Y.; Shenoy, V. B., Ti_3C_2 Mxene as a High Capacity Electrode Material for Metal (Li, Na, K, Ca) Ion Batteries. *ACS Appl. Mater. Interfaces* **2014**, *6*, 11173-11179.
4. Kang, H.; Liu, Y.; Cao, K.; Zhao, Y.; Jiao, L.; Wang, Y.; Yuan, H., Update on Anode Materials for Na-ion Batteries. *J. Mater. Chem. A* **2015**, *3*, 17899-17913.
5. Jian, Z.; Luo, W.; Ji, X., Carbon Electrodes for K-ion Batteries. *J. Am. Chem. Soc.* **2015**, *137*, 11566-11569.
6. Yoo, H. D.; Shterenberg, I.; Gofer, Y.; Gershtinsky, G.; Pour, N.; Aurbach, D., Mg Rechargeable Batteries: An on-going Challenge. *Energy Environ. Sci.* **2013**, *6* 2265-2279.
7. Lin, M. C.; Gong, M.; Lu, B.; Wu, Y.; Wang, D. Y.; Guan, M.; Angell, M.; Chen, C.; Yang, J.; Hwang, B. J.; Dai, H., An Ultrafast Rechargeable Aluminium-ion Battery. *Nature* **2015**, *520*, 325-328.
8. Xue, Y.; Zhang, Q.; Wang, W.; Cao, H.; Yang, Q.; Fu, L., Opening Two-dimensional Materials for Energy Conversion and Storage: A Concept. *Adv. Energy Mater.* **2017**, *7*, 1602684.
9. Zhang, X.; Jin, L.; Dai, X.; Chen, G.; Liu, G., Two-dimensional GaN: An Excellent Electrode Material Providing Fast Ion Diffusion and High Storage Capacity for Li-ion and Na-ion Batteries. *ACS Appl. Mater. Interfaces* **2018**, *10*, 38978-38984.
10. Mukherjee, S.; Banwait, A.; Grixti, S.; Koratkar, N.; Singh, C. V., Adsorption and Diffusion of Lithium and Sodium on Defective Rhenium Disulfide: A First Principles Study. *ACS Appl. Mater. Interfaces* **2018**, *10*, 5373-5384.
11. Xiao, B.; Li, Y.-c.; Yu, X.-f.; Cheng, J.-b., Penta-graphene: A Promising Anode Material as the Li/Na-ion Battery with Both Extremely High Theoretical Capacity and Fast Charge/Discharge Rate. *ACS Appl. Mater. Interfaces* **2016**, *8*, 35342-35352.
12. Liu, C.-S.; Yang, X.-L.; Liu, J.; Ye, X.-J., Theoretical Prediction of Two-dimensional SnP_3 as a Promising Anode Material for Na-ion Batteries. *ACS Appl. Mater. Interfaces* **2018**, *1*, 3850-3859.
13. Tarascon, J. M.; Armand, M., Issues and Challenges Facing Rechargeable Lithium Batteries. *Nature* **2001**, *414*, 359-367.
14. Yang, E.; Ji, H.; Jung, Y., Two-dimensional Transition Metal Dichalcogenide Monolayers as Promising Sodium Ion Battery Anodes. *J. Phys. Chem. C* **2015**, *119*, 26374-26380.
15. Kang, Y.-J.; Jung, S. C.; Choi, J. W.; Han, Y.-K., Important Role of Functional Groups for Sodium Ion Intercalation in Expanded Graphite. *Chem. Mater.* **2015**, *27*, 5402-5406.
16. Xiang, X.; Zhang, K.; Chen, J., Recent Advances and Prospects of Cathode Materials for Sodium-ion Batteries. *Adv. Mater.* **2015**, *27*, 5343-5364.
17. Zhang, K.; Kim, D.; Hu, Z.; Park, M.; Noh, G.; Yang, Y.; Zhang, J.; Lau, V. W.; Chou, S. L.; Cho, M.; Choi, S. Y.; Kang, Y. M., Manganese Based Layered Oxides with Modulated Electronic and

- Thermodynamic Properties for Sodium Ion Batteries. *Nat. Commun.* **2019**, *10*, 5203.
18. Moreau, P.; Guyomard, D.; Gaubicher, J.; Boucher, F., Structure and Stability of Sodium Intercalated Phases in Olivine FePO₄. *Chem. Mater.* **2010**, *22*, 4126-4128.
19. Kim, S.-W.; Seo, D.-H.; Ma, X.; Ceder, G.; Kang, K., Electrode Materials for Rechargeable Sodium-ion Batteries: Potential Alternatives to Current Lithium-ion Batteries. *Adv. Energy Mater.* **2012**, *2*, 710-721.
20. Lu, Y.; Wang, L.; Cheng, J.; Goodenough, J. B., Prussian Blue: A New Framework of Electrode Materials for Sodium Batteries. *Chem. Commun.* **2012**, *48*, 6544-6546.
21. Kulish, V. V.; Malyi, O. I.; Persson, C.; Wu, P., Phosphorene as an Anode Material for Na-ion Batteries: A First-principles Study. *Phys. Chem. Chem. Phys.* **2015**, *17*, 13921-13928.
22. Mukherjee, S.; Kavalsky, L.; Singh, C. V., Ultrahigh Storage and Fast Diffusion of Na and K in Blue Phosphorene Anodes. *ACS Appl. Mater. Interfaces* **2018**, *10*, 8630-8639.
23. David, L.; Bhandavat, R.; Singh, G. J. A. n., MoS₂/Graphene Composite Paper for Sodium-ion Battery Electrodes. *ACS Nano* **2014**, *8*, 1759-1770.
24. Luo, W.; Wan, J.; Ozdemir, B.; Bao, W.; Chen, Y.; Dai, J.; Lin, H.; Xu, Y.; Gu, F.; Barone, V.; Hu, L., Potassium Ion Batteries with Graphitic Materials. *Nano Lett.* **2015**, *15*, 7671-7677.
25. Xie, Y.; Dall'Agnese, Y.; Naguib, M.; Gogotsi, Y.; Barsoum, M. W.; Zhuang, H. L.; Kent, P. R. J. A. n., Prediction and Characterization of MXene Nanosheet Anodes for Non-lithium-ion Batteries. *ACS Nano* **2014**, *8*, 9606-9615.
26. Li, F.; Qu, Y.; Zhao, M., Germanium Sulfide Nanosheet: A Universal Anode Material for Alkali Metal Ion Batteries. *J. Mater. Chem. A* **2016**, *4*, 8905-8912.
27. Ponrouch, A.; Tchitchekova, D.; Frontera, C.; Bardé, F.; Dompablo, M. E. A.-d.; Palacín, M. R., Assessing Si-based Anodes for Ca-ion Batteries: Electrochemical Decalciation of CaSi₂. *Electrochem. Commun.* **2016**, *66*, 75-78.
28. Wang, Z.; Su, Q.; Shi, J.; Deng, H.; Yin, G. Q.; Guan, J.; Wu, M. P.; Zhou, Y. L.; Lou, H. L.; Fu, Y. Q., Comparison of Tetragonal and Cubic Tin as Anode for Mg Ion Batteries. *ACS Appl. Mater. Interfaces* **2014**, *6*, 6786-6789.
29. Jin, T.; Han, Q.; Jiao, L., Binder-free Electrodes for Advanced Sodium-ion Batteries. *Adv. Mater.* **2019**, e1806304.
30. Mortazavi, B.; Dianat, A.; Cuniberti, G.; Rabczuk, T., Application of Silicene, Germanene and Stanene for Na or Li Ion Storage: A Theoretical Investigation. *Electrochim. Acta* **2016**, *213*, 865-870.
31. Banhart, F.; Kotakoski, J.; Krasheninnikov, A. V., Structural Defects in Graphene. *ACS Nano* **2011**, *5*, 26-41.
32. Datta, D.; Li, J.; Koratkar, N.; Shenoy, V. B., Enhanced Lithiation in Defective Graphene. *Carbon* **2014**, *80*, 305-310.
33. Zhu, J.; Schwingenschlögl, U., Silicene for Na-ion Battery Applications. *2D Mater.* **2016**, *3*, 035012.
34. Xu, S.; Fan, X.; Liu, J.; Jiang, Q.; Zheng, W.; Singh, D. J., Adsorption of Na on Silicene for Potential Anode for Na-ion Batteries. *Electrochim. Acta* **2019**, *297*, 497-503.

35. Wu, L.; Lu, P.; Quhe, R.; Wang, Q.; Yang, C.; Guan, P.; Yang, K., Stanene Nanomeshes as Anode Materials for Na-ion Batteries. *J. Mater. Chem. A* **2018**, *6*, 7933-7941.
36. Cao, Y.; Li, M.; Lu, J.; Liu, J.; Amine, K., Bridging the Academic and Industrial Metrics for Next-generation Practical Batteries. *Nat. Nanotechnol.* **2019**, *14*, 200-207.
37. Jeon, D. H., Wettability in Electrodes and Its Impact on the Performance of Lithium-ion Batteries. *Energy Storage Mater.* **2019**, *18*, 139-147.
38. Zhao, Y.; Li, X.; Liu, J.; Zhang, C.; Wang, Q., A New Anisotropic Dirac Cone Material: A B₂S Honeycomb Monolayer. *J Phys. Chem. Lett.* **2018**, *9*, 1815-1820.
39. Li, P.; Li, Z.; Yang, J., Rational Design of Two-dimensional Anode Materials: B₂S as a Strained Graphene. *J Phys. Chem. Lett.* **2018**, *9*, 4852-4856.
40. Delley, B. J. T. J. o. c. p., An All-electron Numerical Method for Solving the Local Density Functional for Polyatomic Molecules. *J. Chem. Phys.* **1990**, *92*, 508-517.
41. Perdew, J. P.; Burke, K.; Ernzerhof, M. J. P. r. l., Generalized Gradient Approximation Made Simple. *Phys. Rev. Lett.* **1996**, *77*, 3865.
42. Grimme, S., Semiempirical Gga-type Density Functional Constructed with a Long-range Dispersion Correction. *J. Comput. Chem.* **2006**, *27*, 1787-1799.
43. Gulbransen, E., The Transition State Theory of the Formation of Thin Oxide Films on Metals. *Trans. Electrochem. Soc.* **1943** *83*, 301-317.
44. Eyring, H., The Activated Complex in Chemical Reactions. *J. Chem. Phys.* **1935**, *3*, 107-115.
45. Krashennnikov, A. V.; Lehtinen, P. O.; Foster, A. S.; Nieminen, R. M., Bending the Rules: Contrasting Vacancy Energetics and Migration in Graphite and Carbon Nanotubes. *Chem. Phys. Lett.* **2006**, *418*, 132-136.
46. Henkelman, G.; Arnaldsson, A.; Jónsson, H., A Fast and Robust Algorithm for Bader Decomposition of Charge Density. *Comput. Mater. Sci.* **2006**, *36*, 354-360.
47. Zhao, H.; Xu, J.; Yin, D.; Du, Y. J. C. A. E. J., A Minireview of Electrolytes for Batteries with Earth-abundant Metallic Anodes. *Chem. Eur. J.* **2018**, *24*, 18220-18234.
48. Zuo, W.; Li, R.; Zhou, C.; Li, Y.; Xia, J.; Liu, J., Battery-supercapacitor Hybrid Devices: Recent Progress and Future Prospects. *Adv. Sci.* **2017**, *4*, 1600539.
49. Ponrouch, A.; Dedryvère, R.; Monti, D.; Demet, A. E.; Mba, J. M. A.; Croguennec, L.; Masquelier, C.; Johansson, P.; Palacín, M. R. J. E.; Science, E., Towards High Energy Density Sodium Ion Batteries through Electrolyte Optimization. *Energy Environ. Sci.* **2013**, *6*, 2361-2369.
50. Hwang, J. Y.; Myung, S. T.; Sun, Y. K., Sodium-ion Batteries: Present and Future. *Chem. Soc. Rev.* **2017**, *46*, 3529-3614.
51. Zhang, Q.; Mao, J.; Pang, W. K.; Zheng, T.; Sencadas, V.; Chen, Y.; Liu, Y.; Guo, Z., Boosting the Potassium Storage Performance of Alloy-based Anode Materials via Electrolyte Salt Chemistry. *Adv. Energy Mater.* **2018**, *8*, 1703288.
52. Goodenough, J. B., Evolution of Strategies for Modern Rechargeable Batteries. *Accounts Chem. Res.* **2013**, *46*, 1053-1061.
53. Wang, Y. J.; Zhou, K. G.; Yu, G.; Zhong, X.; Zhang, H. L., Partial Oxidized Arsenene: Emerging

- Tunable Direct Bandgap Semiconductor. *Sci. Rep.* **2016**, *6*, 24981.
54. Hu, T.; Hong, J., First-principles Study of Metal Adatom Adsorption on Black Phosphorene. *J. Phys. Chem. C* **2015**, *119*, 8199-8207.
55. Hou, J.; Tu, K.; Chen, Z., Two-dimensional Y₂C Electride: A Promising Anode Material for Na-ion Batteries. *J. Phys. Chem. C* **2016**, *120*, 18473-18478.
56. Yu, T.; Zhao, Z.; Liu, L.; Zhang, S.; Xu, H.; Yang, G., TiC₃ Monolayer with High Specific Capacity for Sodium-ion Batteries. *J. Am. Chem. Soc.* **2018**, *140*, 5962-5968.
57. Samad, A.; Shafique, A.; Shin, Y. H., Adsorption and Diffusion of Mono, Di, and Trivalent Ions on Two-dimensional TiS₂. *Nanotechnology* **2017**, *28*, 175401.
58. Nakada, K.; Ishii, A., Migration of Adatom Adsorption on Graphene Using DFT Calculation. *Solid State Commun.* **2011**, *151*, 13-16.
59. Xiang, P.; Chen, X.; Xiao, B.; Wang, Z. M., Highly Flexible Hydrogen Boride Monolayers as Potassium-ion Battery Anodes for Wearable Electronics. *ACS Appl. Mater. Interfaces* **2019**, *11*, 8115-8125.
60. Wang, K.; Jin, Y.; Sun, S.; Huang, Y.; Peng, J.; Luo, J.; Zhang, Q.; Qiu, Y.; Fang, C.; Han, J., Low-cost and High-performance Hard Carbon Anode Materials for Sodium-Ion Batteries. *ACS Omega* **2017**, *2*, 1687-1695.
61. Kuhne, M.; Paolucci, F.; Popovic, J.; Ostrovsky, P. M.; Maier, J.; Smet, J. H., Ultrafast Lithium Diffusion in Bilayer Graphene. *Nat. Nanotechnol.* **2017**, *12*, 895-900.
62. Jiang, H. R.; Lu, Z.; Wu, M. C.; Ciucci, F.; Zhao, T. S., Borophene: A Promising Anode Material Offering High Specific Capacity and High Rate Capability for Lithium-ion Batteries. *Nano Energy* **2016**, *23*, 97-104.
63. Jena, N. K.; Araujo, R. B.; Shukla, V.; Ahuja, R., Borophane as a Benchmate of Graphene: A Potential 2D Material for Anode of Li and Na-ion Batteries. *ACS Appl. Mater. Interfaces* **2017**, *9*, 16148-16158.
64. Shukla, V.; Araujo, R. B.; Jena, N. K.; Ahuja, R., Borophene's Tryst with Stability: Exploring 2D Hydrogen Boride as an Electrode for Rechargeable Batteries. *Phys. Chem. Chem. Phys.* **2018**, *20*, 22008-22016.
65. Shukla, V.; Araujo, R. B.; Jena, N. K.; Ahuja, R., The Curious Case of Two Dimensional Si₂BN: A High-capacity Battery Anode Material. *Nano Energy* **2017**, *41*, 251-260.
66. Xiang, P.; Chen, X.; Zhang, W.; Li, J.; Xiao, B.; Li, L.; Deng, K., Metallic Borophene Polytypes as Lightweight Anode Materials for Non-lithium-ion Batteries. *Phys. Chem. Chem. Phys.* **2017**, *19*, 24945-24954.
67. Makaremi, M.; Mortazavi, B.; Singh, C. V., 2D Hydrogenated Graphene-like Borophene as a High Capacity Anode Material for Improved Li/Na Ion Batteries: A First Principles Study. *Mater. Today Energy* **2018**, *8*, 22-28.
68. Ge, P.; Foulletier, M. J. S. S. I., Electrochemical Intercalation of Sodium in Graphite. *Solid State Ionics* **1988**, *28*, 1172-1175.
69. Mortazavi, M.; Wang, C.; Deng, J.; Shenoy, V. B.; Medhekar, N. V., Ab Initio Characterization

of Layered MoS_2 as Anode for Sodium-Ion Batteries. *J. Power Sources* **2014**, *268*, 279-286.

70. Ren, X.; Zhao, Q.; McCulloch, W. D.; Wu, Y., MoS_2 as a Long-life Host Material for Potassium Ion Intercalation. *Nano Res.* **2017**, *10*, 1313-1321.

71. Sun, Q.; Dai, Y.; Ma, Y.; Jing, T.; Wei, W.; Huang, B., Ab Initio Prediction and Characterization of Mo_2C Monolayer as Anodes for Lithium-Ion and Sodium-ion Batteries. *J. Phys. Chem. Lett.* **2016**, *7*, 937-943.

72. Putungan, D. B.; Lin, S. H.; Kuo, J. L., Metallic VS_2 Monolayer Polytypes as Potential Sodium-ion Battery Anode via ab Initio Random Structure Searching. *ACS Appl. Mater. Interfaces* **2016**, *8*, 18754-18762.

73. Wang, D. S.; Liu, Y. H.; Meng, X.; Wei, Y. J.; Zhao, Y. Y.; Pang, Q.; Chen, G., Two-Dimensional VS_2 Monolayers as Potential Anode Materials for Lithium-Ion Batteries and Beyond: First-Principles Calculations. *J. Mater. Chem. A* **2017**, *5*, 21370-21377.

74. Liu, X.; Wen, Y.; Chen, Z.; Shan, B.; Chen, R., A First-Principles Study of Sodium Adsorption and Diffusion on Phosphorene. *Phys. Chem. Chem. Phys.* **2015**, *17*, 16398-16404.

75. Kulish, V. V.; Malyi, O. I.; Persson, C.; Wu, P., Phosphorene as an Anode Material for Na-Ion Batteries: A First-Principles Study. *Phys. Chem. Chem. Phys.* **2015**, *17*, 13921-13928.

76. Mannix, A. J.; Zhang, Z.; Guisinger, N. P.; Yakobson, B. I.; Hersam, M. C., Borophene as a Prototype for Synthetic 2D Materials Development. *Nat. Nanotechnol.* **2018**, *13*, 444-450.

77. Li, W.; Kong, L.; Chen, C.; Gou, J.; Sheng, S.; Zhang, W.; Li, H.; Chen, L.; Cheng, P.; Wu, K., Experimental Realization of Honeycomb Borophene. *Sci. Bull.* **2018**, *63*, 282-286.

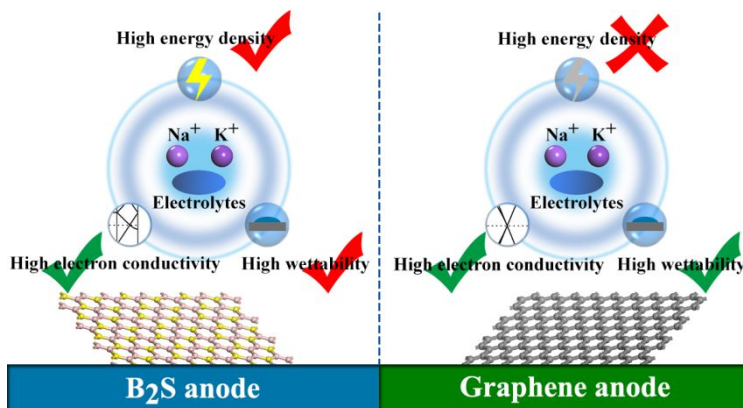


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